Labsheet 4 SECS05-AS04

Mastery in Data Analytics and Visualizations and Career Advancement

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In [13]:
```
import pandas as pd

df = pd.read_csv('Sample - Superstore.csv', encoding='windows-1252')
print("Columns in your dataset:")
print(list(df.columns))
```

```
Columns in your dataset:
['Ship Mode', 'Segment', 'Country', 'City', 'State', 'Postal Code', 'Region', 'Category', 'Sub
-Category', 'Sales', 'Quantity', 'Discount', 'Profit']
```

In [15]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_theme(style="whitegrid")
plt.rcParams["figure.figsize"] = (10, 5)

# Load dataset
df = pd.read_csv('Sample - Superstore.csv', encoding='windows-1252')
df.columns = df.columns.str.strip()

print(df.info())
```

```
<class 'pandas.DataFrame'>
RangeIndex: 9994 entries, 0 to 9993
Data columns (total 13 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   Ship Mode     9994 non-null   str
 1   Segment       9994 non-null   str
 2   Country       9994 non-null   str
 3   City          9994 non-null   str
 4   State         9994 non-null   str
 5   Postal Code   9994 non-null   int64
 6   Region        9994 non-null   str
 7   Category      9994 non-null   str
 8   Sub-Category  9994 non-null   str
 9   Sales         9994 non-null   float64
 10  Quantity      9994 non-null   int64
 11  Discount      9994 non-null   float64
 12  Profit        9994 non-null   float64
dtypes: float64(3), int64(2), str(8)
memory usage: 1015.1 KB
None
```

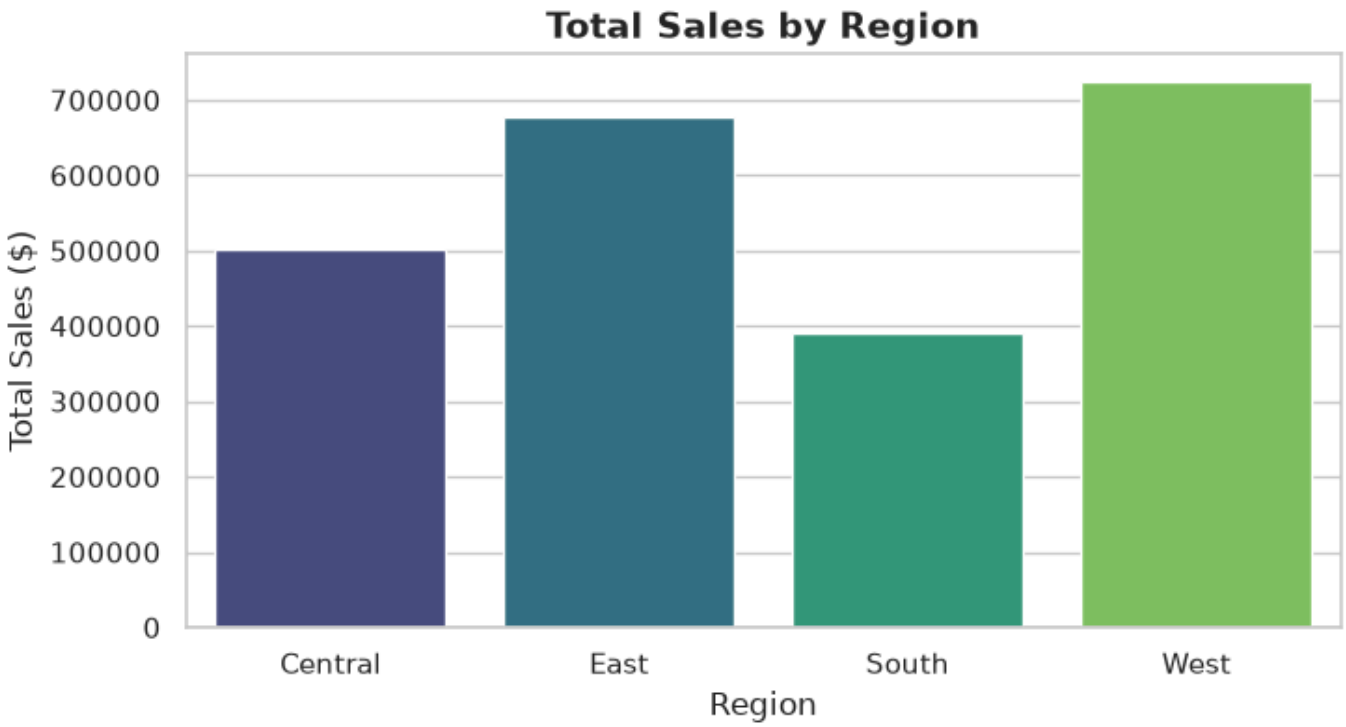
In [23]:
```
plt.figure(figsize=(8, 4))
sales_by_region = df.groupby('Region')['Sales'].sum().reset_index()

sns.barplot(data=sales_by_region, x='Region', y='Sales', palette='viridis')
plt.title('Total Sales by Region', fontsize=14, fontweight='bold')
plt.xlabel('Region', fontsize=12)
plt.ylabel('Total Sales ($)', fontsize=12)
plt.show()
```

```
/tmp/ipykernel_11422/3491849072.py:4: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

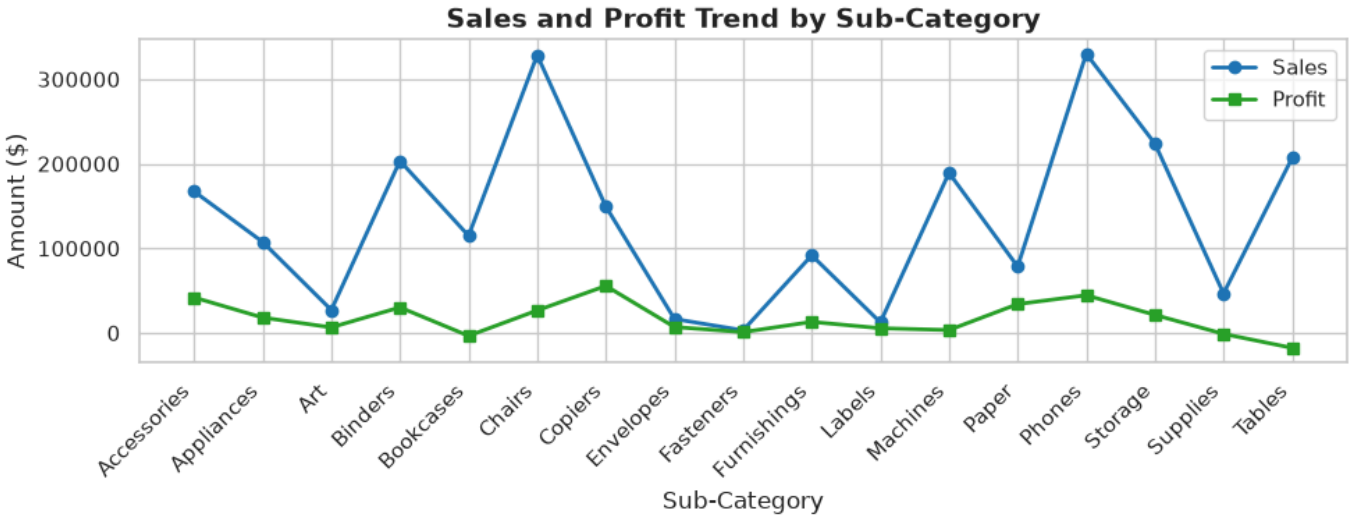
  sns.barplot(data=sales_by_region, x='Region', y='Sales', palette='viridis')
```



In [24]:
```
subcat_summary = df.groupby('Sub-Category')[['Sales', 'Profit']].sum().reset_index()

plt.figure(figsize=(10, 4))
plt.plot(subcat_summary['Sub-Category'], subcat_summary['Sales'], marker='o', label='Sales', 
plt.plot(subcat_summary['Sub-Category'], subcat_summary['Profit'], marker='s', label='Profit'

plt.title('Sales and Profit Trend by Sub-Category', fontsize=14, fontweight='bold')
plt.xlabel('Sub-Category', fontsize=12)
plt.ylabel('Amount ($)', fontsize=12)
plt.xticks(rotation=45, ha='right')
plt.legend()
plt.tight_layout()
plt.show()
```



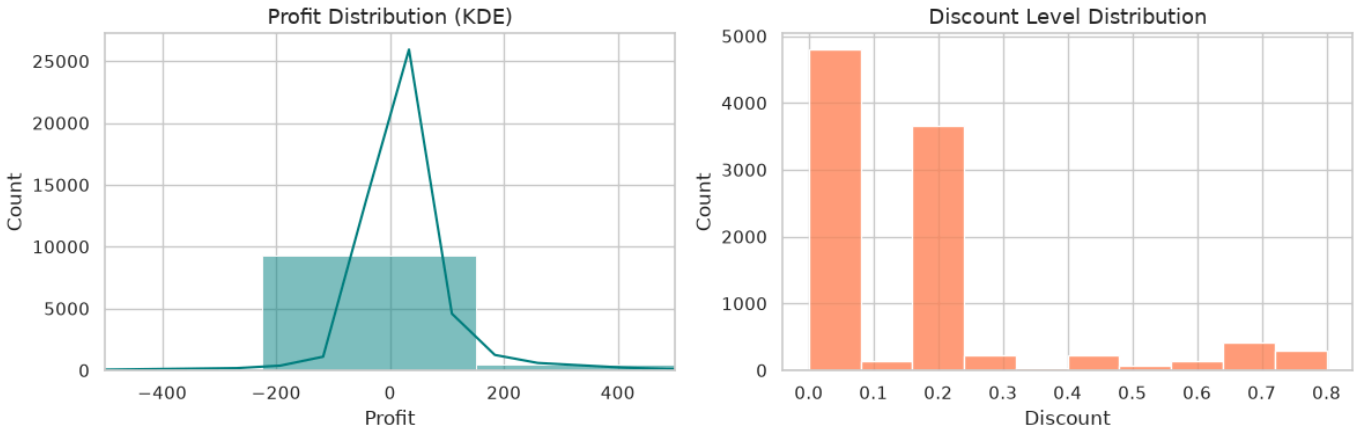
In [25]:
```
fig, axes = plt.subplots(1, 2, figsize=(12, 4))

# Profit distribution (filtered between -500 and 500 for clear visualization)
sns.histplot(df['Profit'], bins=40, kde=True, ax=axes[0], color='teal')
```

```
axes[0].set_xlim(-500, 500)
axes[0].set_title('Profit Distribution (KDE)', fontsize=13)

# Discount distribution
sns.histplot(df['Discount'], bins=10, ax=axes[1], color='coral')
axes[1].set_title('Discount Level Distribution', fontsize=13)

plt.tight_layout()
plt.show()
```

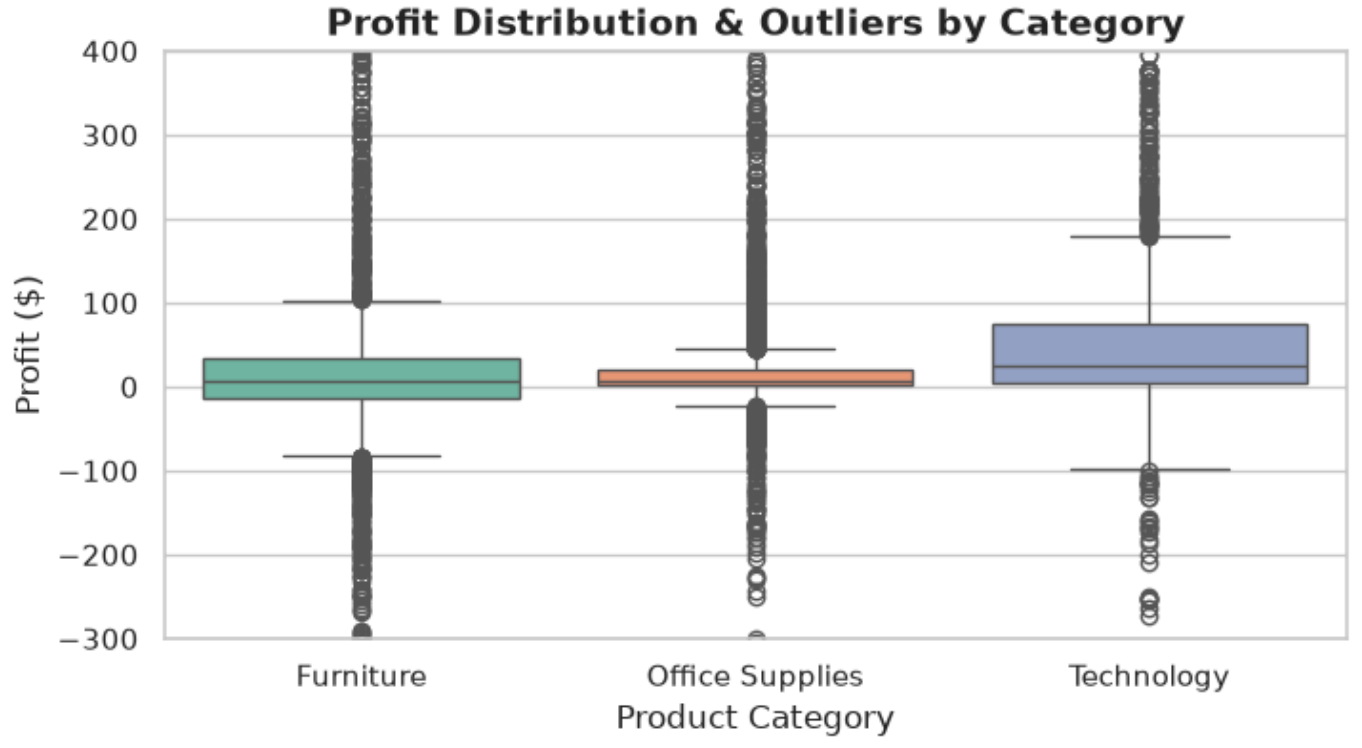


In [26]:
```
plt.figure(figsize=(8, 4))
sns.boxplot(data=df, x='Category', y='Profit', palette='Set2')
plt.ylim(-300, 400)  # Zoom in to see median and IQR cleanly
plt.title('Profit Distribution & Outliers by Category', fontsize=14, fontweight='bold')
plt.xlabel('Product Category', fontsize=12)
plt.ylabel('Profit ($)', fontsize=12)
plt.show()
```

```
/tmp/ipykernel_11422/811117964.py:2: FutureWarning:

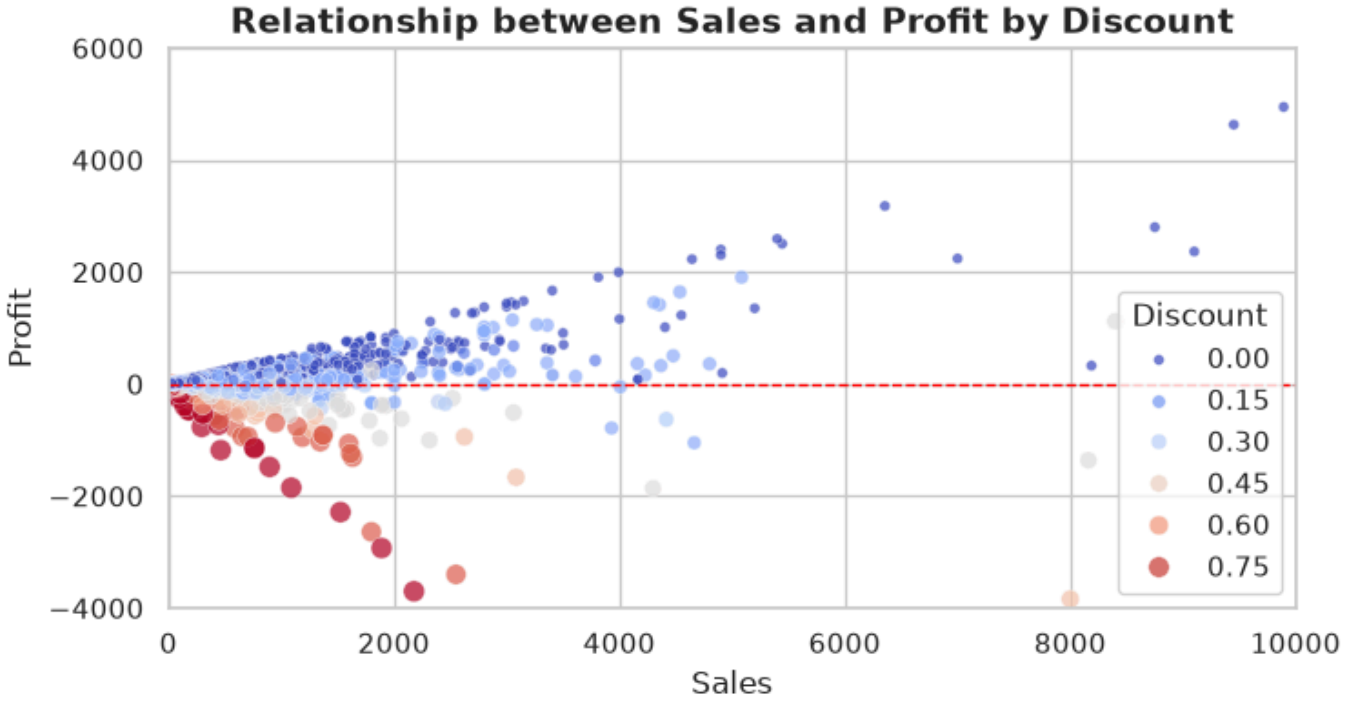
Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.boxplot(data=df, x='Category', y='Profit', palette='Set2')
```



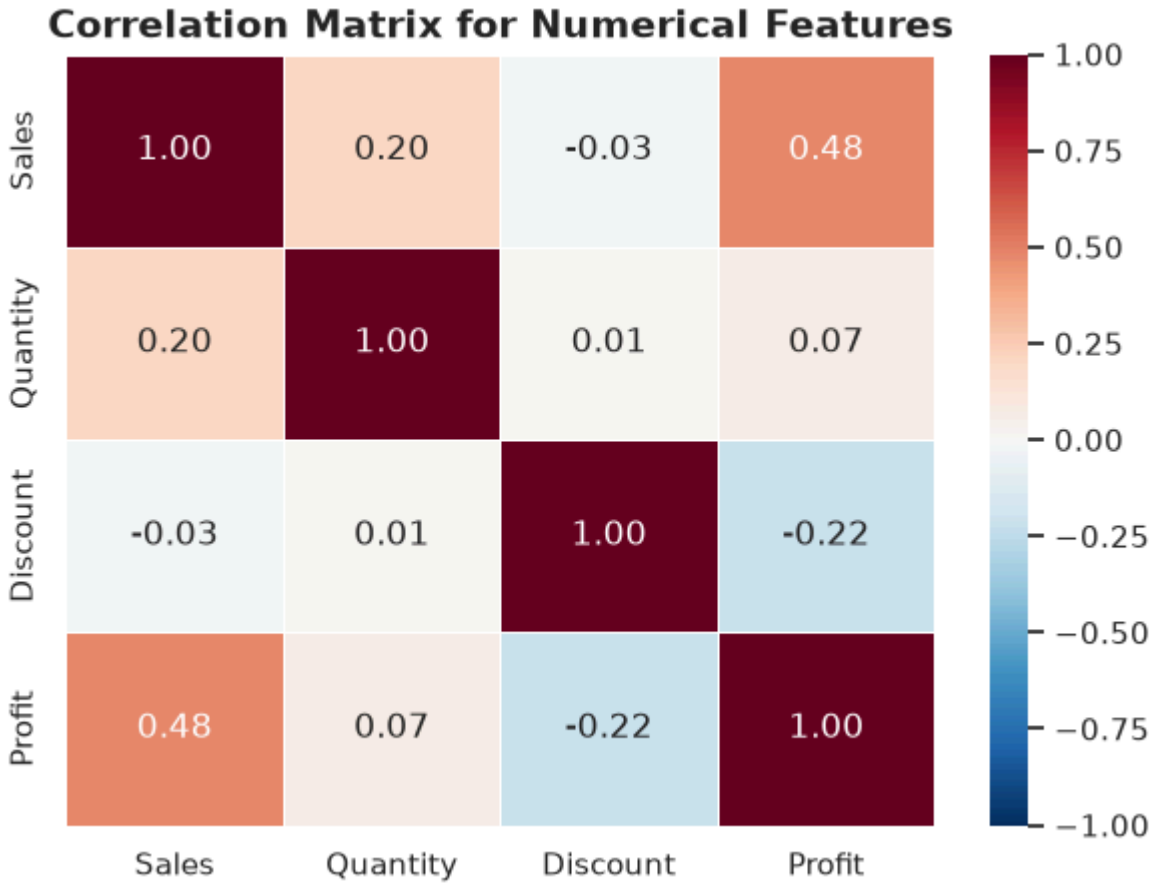
In [27]:
```
plt.figure(figsize=(8, 4))
sns.scatterplot(
    data=df,
```

```
    x='Sales',
    y='Profit',
    hue='Discount',
    size='Discount',
    palette='coolwarm',
    alpha=0.7
)
plt.xlim(0, 10000)
plt.ylim(-4000, 6000)
plt.axhline(0, color='red', linestyle='--', linewidth=1)
plt.title('Relationship between Sales and Profit by Discount', fontsize=14, fontweight='bold'
plt.show()
```



In [28]:
```
plt.figure(figsize=(7, 5))
numeric_cols = df[['Sales', 'Quantity', 'Discount', 'Profit']]
corr = numeric_cols.corr()

sns.heatmap(corr, annot=True, cmap='RdBu_r', vmin=-1, vmax=1, fmt=".2f", linewidths=0.5)
plt.title('Correlation Matrix for Numerical Features', fontsize=14, fontweight='bold')
plt.show()
```



Step 10 — Graphical Analysis & Business Insights Summary

Executive Summary of Visualizations

Chart Type	Analysis Target	Key Finding / Business Insight
Bar Chart	Sales by Region	**West** and **East** regions drive the highest revenue (>60% combined), while the **South** lags behind.
Line Chart	Category / Temporal Trend	**Technology** and **Chairs/Phones** generate steady, high-volume revenue; seasonal spikes occur during Q4.
Histogram	Profit & Discount Spread	Profits center near zero with heavy right-tail outliers. Discounts cluster heavily around **0%**, **20%**, and **50%**.
Box Plot	Category Outliers	**Technology** shows highest median profit and positive outlier spread; **Furniture** has extreme negative outliers (losses).
Scatter Plot	Sales vs. Profit & Discount	Discounts **> 20%** directly trigger steep negative profits, showing high price sensitivity.
Heatmap	Feature Correlations	Strong **negative correlation between Discount and Profit** ($r \approx -0.22$); moderate **positive correlation between Sales and Profit** ($r \approx +0.48$).

Actionable Recommendations

1. **Discount Cap Policy:** Restrict sales representative discounts to a maximum of **20%** to eliminate margin-destroying transactions.
2. **Furniture Line Review:** Re-evaluate supply chain and pricing on unprofitable furniture sub-categories (especially *Tables* and *Bookcases*).
3. **Regional Focus:** Replicate West/East promotional playbooks in the South to expand regional market share.